

2026 Risk Table Summary for Eastern Bering Sea Snow Crab

The following is a synthesis and interpretation of the most recent ecosystem and socioeconomic information available for eastern Bering Sea (EBS) snow crab from the Ecosystem and Socioeconomic Profile (ESP, Fedewa et al., 2026), the AFSC Spring Preview of Ecosystem and Economic Conditions Workshop (AFSC Spring PEEC 2026), and the Eastern Bering Sea Ecosystem Status Report (ESR; Siddon, 2026, in press). This information may be helpful for evaluating risk table score levels and is organized below by the proposed risk table categories.

Using Dynamic Structural Equation Modeling (DSEM), we identified **spring sea ice extent**, **juvenile snow crab energetic condition**, **cold-water habitat availability**, and **juvenile snow crab temperature of occupancy** as ESP indicators with statistically supported causal relationships with snow crab recruitment. Specifically, declines in temperatures occupied by juvenile snow crab, and elevated sea ice extent, energetic condition and cold-water habitat availability are expected to drive increases in snow crab recruitment. These indicators are emphasized in **BOLD** below.

Current Year Population Dynamics Considerations:

Risk Level 1: Normal

- **Juvenile snow crab energetic condition** rebounded following the population collapse and remains elevated in 2026, well above lab-derived starvation thresholds (ESP: Fedewa et al., 2026).
- **Juvenile snow crab occupied temperatures** were $< 0^{\circ}\text{C}$ in 2026, indicating conditions suitable for high survival, recruitment and stock recovery (ESP: Fedewa et al., 2025).
- Bitter crab disease and Pacific cod predation indicators that represent proximate mechanisms for increased mortality of juvenile snow crab remained below average in 2026 (ESP: Fedewa et al., 2025; *Pacific cod consumption estimate for 2026 still in progress*).
- The male snow crab size at 50% maturity remains near average in 2026, although this indicator has been trending down for the past three decades. Mature female size at maturity declined in 2026 following several cohorts of large mature females observed in 2024 and 2025 (ESP: Fedewa et al., 2026).
- Record-high mature female abundance in 2026 resulted in a strongly female-skewed operational sex ratio (85 mature females for every one $>95\text{mm}$ large male), although only $\sim 3\%$ of mature females with empty clutches suggests high reproductive potential despite depressed abundances of large male snow crab (ESP: Fedewa et al., 2026).
- Mature male snow crab center of abundance and area occupied remain near average in 2026 despite relatively cold summer temperatures (ESP: Fedewa et al., 2026).
- A pulse of $\sim 10\text{mm}$ CW snow crab prey in Pacific cod stomachs in summer 2025 may be indicative of an upcoming recruitment event (Aydin 2026, in AFSC Spring PEEC 2026; Aydin 2026, in Siddon 2026, in press).

Current Year Ecosystem Considerations:

Risk Level 1: Normal

Ecosystem indicators are organized into several categories to capture the scope of considerations available in the ESP and ESR reports:

Distribution:

- The along-shelf winds in June, July, and August 2025 were weakly favorable to on-shelf Ekman transport. Onshelf transport may have retained pelagic crab stages over the shelf and over suitable benthic habitat in 2025.
- March 2026 along-shelf winds drove strong offshore transport and cross-shelf winds that drove transport to the northwest (Hennon 2026, in AFSC Spring PEEC 2026; Hennon 2026, in Siddon 2026, in press). Spring offshore transport may have dispersed pelagic crab larval stages away from suitable benthic habitat during peak hatch in April-May.

Environmental Processes:

- During winter 2025-2026 (Dec 2025 - Mar 2026), the Aleutian Low Pressure Index was weak, which tends to result in fewer storms and weaker winds (Overland & Wang 2026, in AFSC Spring PEEC 2026; Overland & Wang 2026, in Siddon 2026, in press). Sea ice extent was near average through February 2026 (ice advanced over residual warm SSTs), then winds from the Arctic during March/April led to record ice extent (for 6 weeks), but ice was thin (Lemagie 2026, in AFSC Spring PEEC 2026; Lemagie 2026, in Siddon 2026, in press).
- The Arctic Oscillation was in a slightly negative state during winter/spring 2026, which is expected to favor snow crab recruitment (ESP: Fedewa et al., 2026).
- **Spring sea ice extent** increased substantially in 2026 and is above the time series mean. 56% of the EBS and NBS survey was covered with sea ice in March-April (ESP: Fedewa et al., 2026).
- **The spatial extent of cold-water habitat $\leq 0^{\circ}\text{C}$** was near average in 2026 and the spatial extent of the cold pool ($\leq 2^{\circ}\text{C}$) was above average in 2026. While waters $\leq 0^{\circ}\text{C}$ appear to be more important for snow crab, both indicators suggest optimal availability of cold-water habitat for highly stenothermic juvenile snow crab (ESP: Fedewa et al., 2026).
- The NMME ensemble forecasts as of August 21, 2026 through November 2026 show cool SST anomalies over much of the SEBS ($-0.5 - 0^{\circ}\text{C}$) and warm anomalies over the NBS ($0.25 - 1^{\circ}\text{C}$), with warmer anomalies near shore (Lemagie 2026, in Siddon 2026, in press; *forecast will be updated for final ESR*).
- Taken together, thermal conditions over the EBS shelf from winter 2025 through summer 2026 have been cool with an extensive cold pool in summer 2026. High sea ice extent and cold pool extent are favorable for snow crab recruitment and productivity.

Prey:

- Diatom abundance anomalies, based on the Continuous Plankton Recorder, remained positive from 2023 through 2025 (Ostle & Batten 2026, in AFSC Spring PEEC 2026; Ostle & Batten 2026, in Siddon 2026, in press), indicating above-average feeding conditions for pelagic crab stages.
- Satellite chlorophyll data showed an early spring bloom associated with ice retreat. Spatially, large areas with ice associated blooms occurred over the shelf; ice associated blooms tend to lead to higher deposition of algae to the seafloor, indicating good feeding conditions for benthic crab stages (Nielsen 2026, in AFSC Spring PEEC 2026; Gann & Callahan 2026, in Siddon 2026, in press). Summer 2026 chlorophyll-a concentration on the north-middle shelf remains near average (ESP: Fedewa et al., 2026).
- Benthic prey density in core snow crab habitat was well above average in 2025 (ESP: Fedewa et al., 2026; indicator update for 2026 in progress).
- Taken together, feeding conditions for pelagic crab stages have been favorable in 2025 and 2026. Measurements of prey for benthic crab stages were mixed in 2025.

Competitors:

- Over the southern shelf, motile epifauna (e.g., sea stars, brittle stars, crabs) biomass increased from 2024 to 2025 and is more than two standard deviations above the long term mean (Whitehouse 2026, in Siddon 2026, in press). Nearly all functional groups within this guild increased from 2024 to 2025, including commercial crabs. Benthic forager (i.e., small-mouthed flatfish) biomass increased slightly from 2024 to 2025 but remained one standard deviation below the time series mean, suggesting competition for prey resources remained low in 2025 (Whitehouse 2026, in Siddon 2026, in press).
- The presence of hybrid Tanner-snow crab prey in Pacific cod diets remained high between 2023-2025, potentially indicating increased competition from hybrids (Aydin 2026, in AFSC Spring PEEC 2026; Aydin 2026, in Siddon 2026, in press)
- Taken together, competition for space continues to be high, but competition for prey resources remains low.

Predators:

- The biomass of Pacific cod during the standard bottom trawl survey decreased 10% from 2024 to 2025 (Whitehouse 2026, in Siddon 2026, in press), indicating a reduction in predation pressure in 2025.
- Pacific cod consumption of snow and tanner crab in 2025 was similar to 2023-2024 (Aydin 2026, in AFSC Spring PEEC 2026; Aydin 2026, in Siddon 2026, in press; ESP: Fedewa et al., 2026).

Current Year Fishery-informed Stock Considerations:

Risk Level 1: Normal

Considerations are from updated fishery performance indicators reported in the ESP (Fedewa et al., 2026) and results from the ABSC Skipper Survey regarding the 2025/2026 snow crab fishery.

- CPUE of retained crab in the 2025/26 fishery declined relative to the prior 2024/25 season (219 crab/potlift to 147 crab/potlift), and remains below the long term average (ESP: Fedewa et al., 2026).
- Total effort in the fishery, as measured by number of active vessels (36) and total potlifts (83.4 thousand), increased relative to the previous year fishery but remains below the post-rationalization average (ESP: Fedewa et al., 2026).
- The center of distribution of fishing activity in the 2025/26 fishery shifted substantially to the SE at 57.45 degrees North latitude, relative to 2024/25. This is consistent with ~60% of skippers in the ABSC Skipper Survey noting that they fished further south and/or changed fishing locations more frequently. Skippers reported that this behavioral change was primarily due to weather and sea ice, but many skippers also noted heavy sorting of small males and females as a motivator (ESP: Fedewa et al., 2026).
- Skipper survey responses varied when asked what drove the location of their fishing this season but many skippers attributed the decision to fuel consideration during a low TAC year and finding the closest grounds with decent catch (ESP: Fedewa et al., 2026).
- ~50% of skippers who responded to ABSC's Skipper Survey reported a >10% increase in perceived abundance of industry-preferred males relative to the previous open season (ESP: Fedewa et al., 2026).
- 47% of skippers noted slightly more to a lot more hybrid crab than the previous season, and the majority of skippers felt confident that their crew could accurately identify and sort hybrids (ESP: Fedewa et al., 2026).

References

AFSC Spring PEEC 2026. Preview of Ecosystem and Economic Conditions Workshop. May 25-28, 2026. NOAA Alaska Fisheries Science Center, Seattle, WA, USA.

(<https://docs.google.com/document/d/1XUsvHh2frm7YD3Cemfa5omS0IEzBeU89y3IlnK4Tx7A/edit?tab=t.0#heading=h.95r7ro4r01pj>)

Fedewa, E., and Shotwell, K. 2026. Appendix A. Ecosystem and Socioeconomic Profile of the eastern Bering Sea snow crab stock - Update. *In*: Adams and Szuwalski 2026. An Assessment for eastern Bering Sea snow crab. North Pacific Fishery Management Council, Anchorage, AK.

Siddon, E. 2026 (in press). Ecosystem Status Report 2026: Eastern Bering Sea, Stock Assessment and Fishery Evaluation Report, North Pacific Fishery Management Council, 1007 West 3rd Ave., Suite 400, Anchorage, Alaska 99501.

	Assessment-related Considerations	Population Dynamics Considerations	Ecosystem Considerations	Fishery-informed Stock Considerations
Long-term/Persistent Considerations	Assessment Author to complete	Assessment Author to complete		
Current Year	Assessment Author to complete	Level 1: Normal Stock-specific indicators related to natural mortality, growth and recruitment suggest no apparent population dynamics concerns, although a highly female-skewed sex ratio combined with directional downward shifts in male maturity and depressed abundances of large males should continue to be monitored.	Level 1: Normal Offshelf transport and away from suitable benthic habitat in spring 2026. Thermal conditions over the shelf from winter 2025 to summer 2026 have been favorable for snow crab, with an extensive cold pool and high sea ice extent. Pelagic feeding conditions have remained favorable since 2023; benthic prey indicators were mixed in 2025. Competition for space remained high, while competition for prey remained low in 2025. Predation pressure from Pacific cod was reduced in 2025. Overall, ecosystem concerns are minor with uncertain impacts on the stock.	Level 1: Normal Fishery-informed indicators generally support stable stock condition. Shifts in fleet behavior appear to be motivated primarily by weather, sorting, fuel costs and TAC considerations rather than stock health or distribution. Many skippers noted increased hybrid abundance during the 2025/26 fishery, consistent with the dramatic increase in hybrid abundances observed on the 2025 EBS bottom trawl survey.

